

Active Beauty

Agefinity™

The 3D Y reshaper

Crafted by white and green technology



Givaudan

engage your senses

*‘The first feminine feature
that goes, with advancing
age, is the neck.’*

Gloria SWANSON – Silent films american actress

The Y-zone (face and neck)

Requires specific care with age

- Skin undergoes **progressive structural** and **functional** decline.
- Epidermis decreases in **thickness** by more than 6% every decade (especially lower cheeks, neck and upper part of the chest).¹
- Epidermal **turnover** is reduced.²
- Active melanocytes are reduced from -8 to -20% every decade leading **to uneven skin tones**.³
- Loss of **molecular integrity** of the reticular dermis leading to **rigidity**, decrease of torsion extensibility and **elasticity**

1- Waller JM, Maibech HI – Age and skin structure and function, a quantitative approach(I) blood flow, pH, thickness, and ultrasound echogenicity. Skin res Technol 2005, 11:221

2-Suter-Widmer J, Elsner P: Age and irritation. The irritant contact dermatitis syndrome. 1996 257:265

3- Phillips Tkanj L Clinical manifestation of skin ageing. The effect of skin ageing in oral mucosa and skin. 1994. 25-40.



The neck

shows our age before our face

Turkey neck

is a rather unglamorous term for **wrinkled, sagging skin** on the neck, which is a common side effect of ageing. **Vertical lines** occur when the neck muscles weaken and skin loses **elasticity**, and ability to stretch and stay tight.



Crepey neck

Is due to a **decrease in collagen**. Collagen production decreases over time upon ageing, and is also damaged by environmental hazards, such as UV radiations, pollution etc. **Horizontal wrinkles** usually show up as **dry, crepey** skin.



The neck is even more under pressure with new technologies

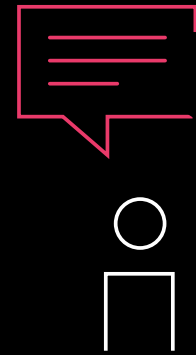
Tech neck

Dermatologists and other skin care experts believe that constantly looking downward at our smartphones could lead to premature skin sagging and wrinkles.



Market understanding

Consumers & Market insight



Consumers & Market Insight

Givaudan Active Beauty expertise

OBJECTIVES

Understand consumer **perception** of skin ageing.

METHODOLOGY

On line survey and data collection



DESIGN

- February 2019
- Sample size:
1230 respondents
- G-Source panel



TARGET

- Covered 6 countries
- Women; aged from 18 to 50 y.o.



ANALYSIS

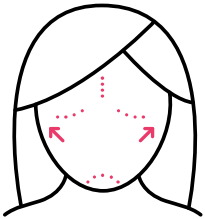
- 300k+ data on 88 questions & demographics
- 10k+ verbatim analysed

When age matters...

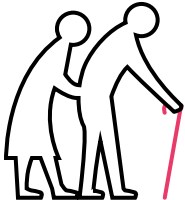
Some key numbers & facts



80% of people agree that 'ageing skin makes a woman look older than her age' (up to 86% in Korea).



Lines & wrinkles, sagging skin and age spots are the top 3 most important signs of ageing (respectively for 78%, 75% and 68% of people).

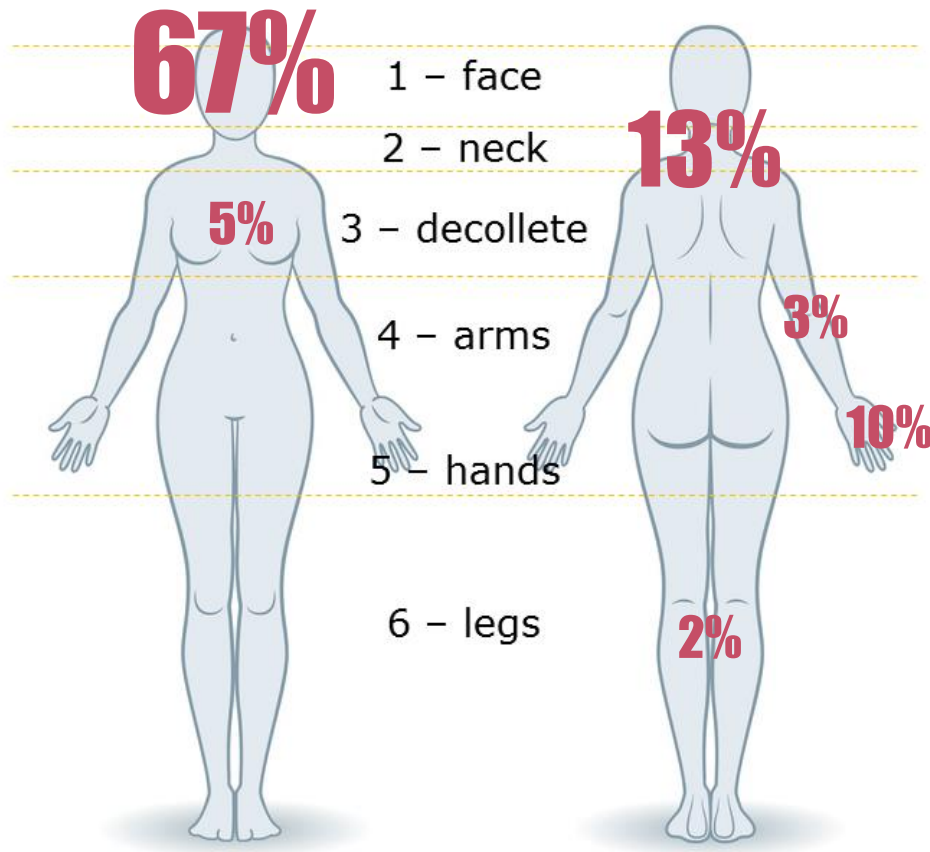


75% of people find that ageing skin has an impact on their life (up to 92% in China).

Source: Givaudan on line survey over 6 countries (USA, Brazil, France, UK, Korea, China), 1230 respondents (women from 18 to 50 years old), February 2019

Face above all but neck is also critical...

Question: When you come across a woman that you do not know, which of the zones would you focus on to guess her age?



Neck is the
#2 focus area
to guess a woman's
age.

Face is obviously #1.

Source: Givaudan on line survey over 6 countries (USA, Brazil, France, UK, Korea, China), 1230 respondents (women from 18 to 50 years old), February 2019

“Tech Neck” is a very common phenomenon globally

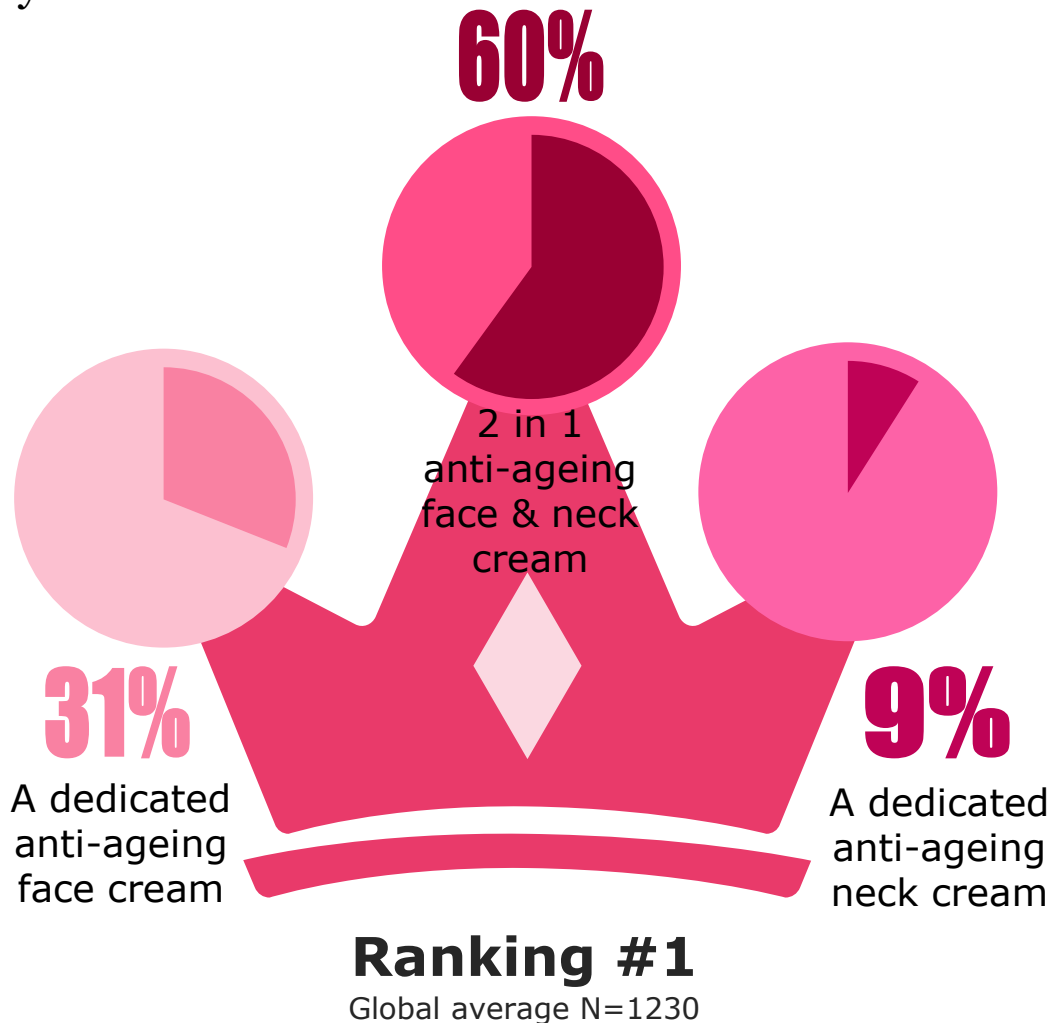
Question: How much do you feel concerned by the "Tech Neck" and the wrinkles? (Extremely + very concerned)



Source: Givaudan on line survey over 6 countries (USA, Brazil, France, UK, Korea, China), 1230 respondents (women from 18 to 50 years old), February 2019

2 in 1 products are more attractive to consumers globally

Question: Please rank the following products according to the level it attracts you.

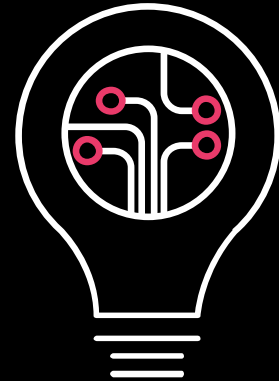


As both face and neck are important zones for “ageing”, 2 in 1 product is more attractive.

In China and Korea, consumers are keener on the dedicated face cream which is perceived more efficient.

Concept of the product

Agefinity™



The reticular dermis

Essential to maintain the Y-zone

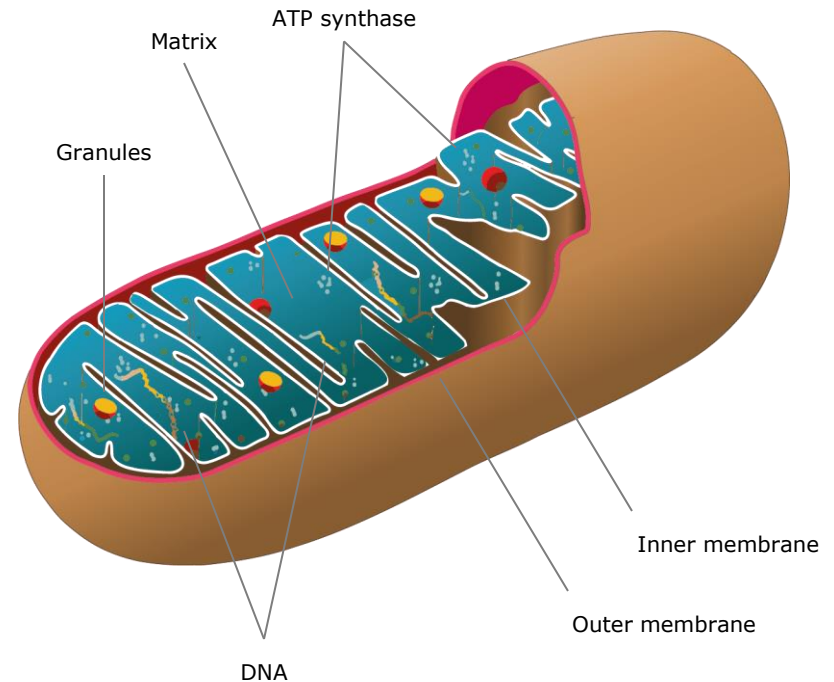
- **Lower layer of the dermis**, found under the papillary dermis.
- Composed of dense irregular connective tissues and **densely packed collagen fibers**.
- It gives the dermis **strength, extensibility, and elasticity**.
- **Orientation of collagen fibres** within the reticular dermis creates lines of tension called Langer's lines, which are essential for wound healing and mechanical properties.



Mitochondria

The power house of the cells

- Unique **organelles** which are the power generators of cells.
- Converting oxygen/nutrients into chemical energy (**ATP**) allowing cells metabolism.
- Support **essential cell functions** (energy powerhouse, protection of DNA, signaling cell reproduction, activation of cell apoptosis, maintenance of cell electrochemical integrity...)



Decline of mitochondrial activity Leads to disorganization of reticular dermis

Age and external stresses damage the mitochondria of dermal fibroblasts:

- Reactive Oxygen Species (**ROS**) production.
- Inadequate **energy** production.
- **Genes expression** pattern of dermal fibroblasts is changed.
- Perturbation of **collagen production**, **inflammation**, disruption of **detoxification** pathway (proteasome) and **oxidation of skin pigments** (lipofuscin).

→ **Disorganisation of the extracellular matrix (ECM), premature skin ageing and apparition of ageing spots.**



Mannose-6-Phosphate

Recharge your skin with energy

- Mannose 6P is a human metabolite involved in the production of Fructose 6P, which is itself involved in glycolysis and mitochondrial activity.

In human cells:

Mannose 6P → Fructose 6P → Energy

- We use a natural source of Mannose and activate it into its high energy equivalent Mannose 6P to deliver a ***clean beauty*** active ingredient.

Mannose + Phosphate $\xrightarrow{\text{enzyme}}$ Mannose 6P
(plant)

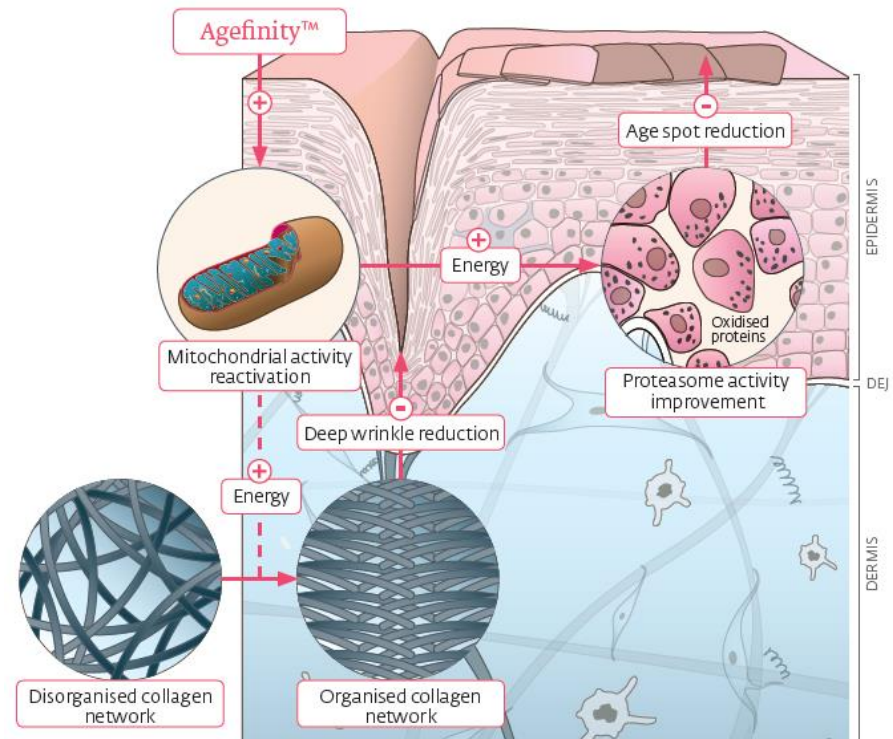


Agefinity™

Mode of action

To replenish the energy of the cells, Active Beauty has crafted **Agefinity™**, a Mannose-6-Phosphate complex, which will:

- Reprogram **mitochondrial energy metabolism** in older cells.
- Promote **skin matrix restructuring**.
- Reinforce **Dermo Epithelial Junction** (DEJ).
- **Reorganise skin matrix** similar to younger skin.
- Significantly **reduce visible age spots**.
- Visibly **reduce crows feet wrinkles**.
- Reduce neck wrinkles to **remodel the Y-shape** of the face.



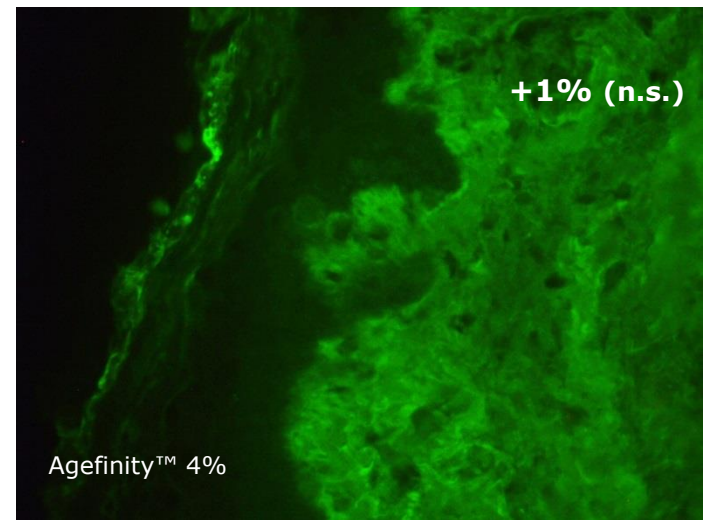
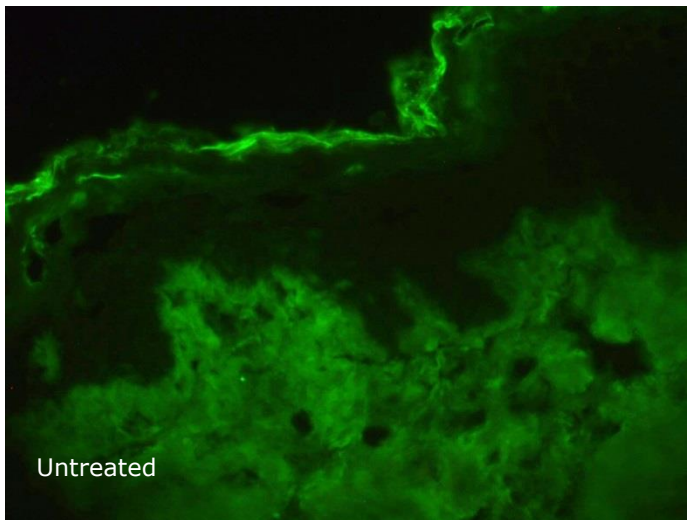
A sugar with no impact on
protein glycation

A sugar with no impact on protein glycation (*ex vivo*)

Protocol:

Skin explants were treated with 4% of Agefinity™ or left untreated for 48 hours.

The glycation (bonding of a sugar molecule to a protein or lipid molecule without enzymatic regulation that damages proteins) was evaluated using specific immunohistochemistry with anti-AGE primary antibody.



n.s. not significant

Results:

Agefinity™ **doesn't induce protein glycation** due to the exposition to sugar.

Reactivating mitochondrial energy metabolism

Action on mitochondrial target (*in vitro*)

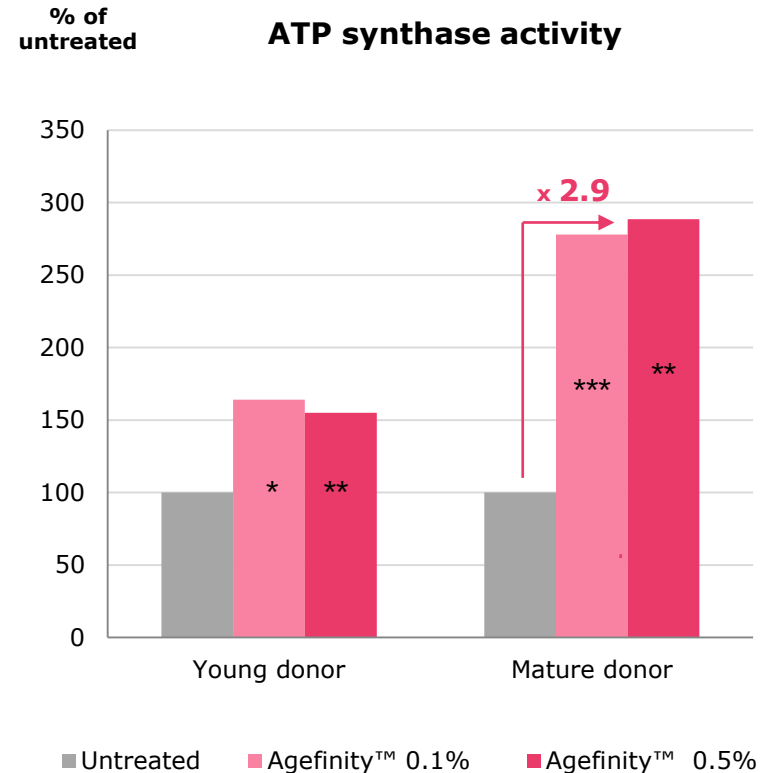
Protocol:

Keratinocytes (NHEK) from a young donor's skin (26 year old) and from an older donor's skin (51 year old), were treated with 0.1% and 0.5% of Agefinity™ for 72 hours or left untreated.

The ATP synthase enzymatic activity was then evaluated using a spectrofluorometer.

Results:

Agefinity™ significantly improves the cells metabolism, with up to 2.9 times **increase of ATP synthase activity**, in older cells.



Student's t-test vs untreated:
*p<0.05 **p<0.01 ***p<0.001

Skin matrix restructuration

Proteomic analysis (*ex vivo*)

Protocol:

Skin explants from a 59 year old Caucasian donor were treated with 4% of Agefinity™ for 7 days or left untreated. The variation of protein expressions were quantified using immunostaining quantification after specific immunohistochemistry.

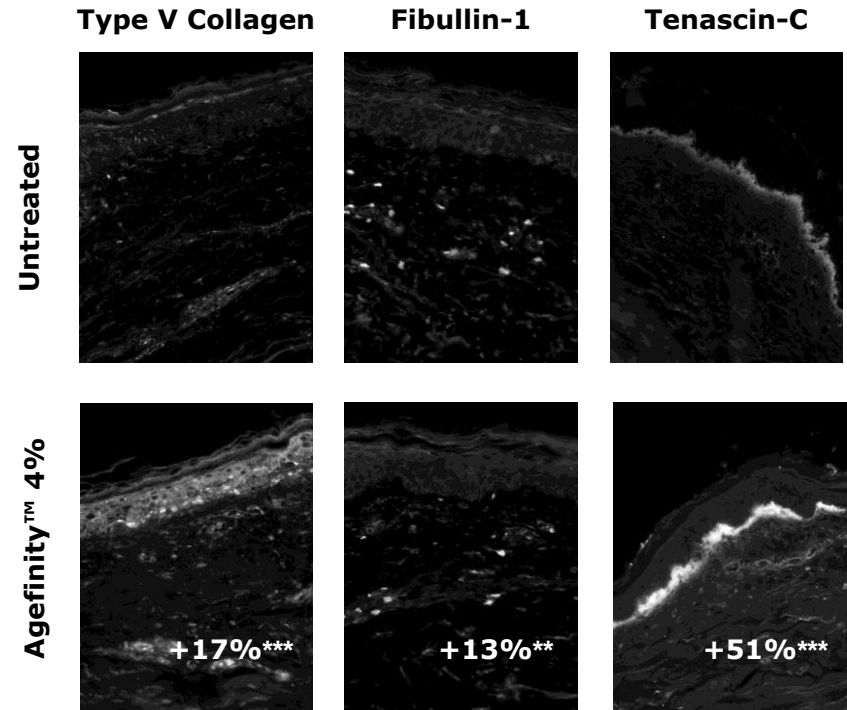
Type V Collagen: Fibrillary collagen, minor component of the dermal matrix which associates with collagen type I

Fibullin-1: component of the extracellular matrix essential to maintain structure of the DEJ

Tenascin-C: matrix protein involved in the cell cohesion at the DEJ located at the basal layer.

Results:

Agefinity™ **stimulates the expression of various proteins involved in dermis structuration.**



** p<0.01 Student's t-test
*** p<0.001 Student's t-test

Morphological analysis (*ex vivo*)

Protocol:

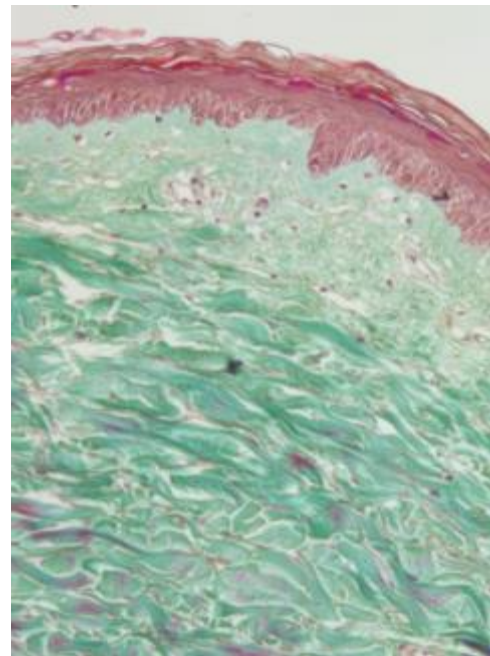
Skin explants from a 59 year old Caucasian donor were treated with 4% of Agefinity™ for 7 days or left untreated. Collagen fibres and muscular tissues were stained using Masson's Trichrome staining.

Results:

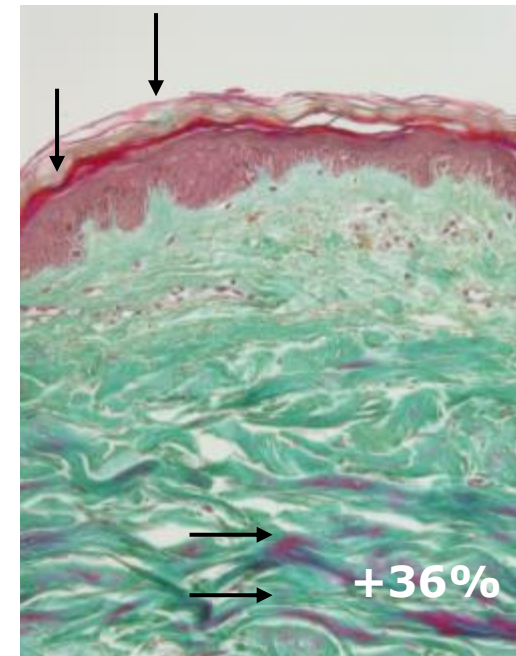
Agefinity™ at 4% visibly **increases the density of reticular dermis** by +36%, the DEJ is better defined and the epidermis is more cohesive.

- Nucleus
- Muscular fibres, erythrocytes, cytoplasm, elastic fibres.
- Collagen fibres

Untreated



Agefinity™ 4%



Atomic Force Microscopy

A unique technique to analyze the reticular dermis

- **Scan the surface** of skin cells or skin sections,
- A sharp tip identifies **and follow the relief of the support** during the scanning.
- This method gives information about **biomechanical** properties and **topography** of the studied support.



Production: Physics Reimagined group (LPS, CNRS Université Paris-Sud) with funding of Labex PALM.
<https://www.youtube.com/watch?v=8gCf1sEn0UU>

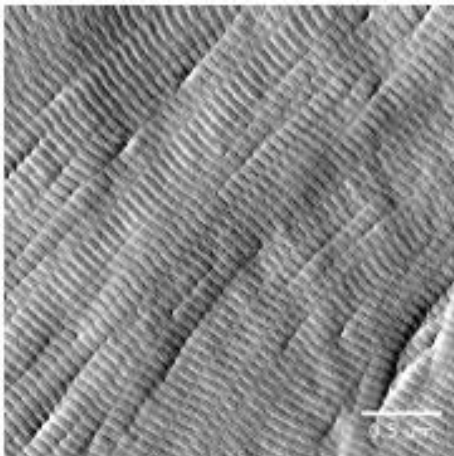
Collagen fibres organisation (*ex vivo*)

Protocol:

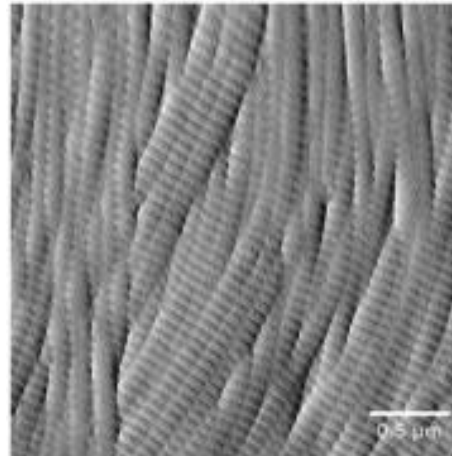
Skin explants from 2 donors (30 and 65 year old Caucasian women) were topically treated every day with 4% of Agefinity™ for 8 days or left untreated. The organisation of collagen fibres into young and mature explants was evaluated **using Atomic Force Microscopy (AFM)** and Small-angle X-ray scattering (SAXS).

This technique (AFM) scans the surface of skin cells or skin sections, using a sharp tip which will identify and follow the relief of the support during the scanning. **This method gives information about biomechanical properties and topography of the studied support.**

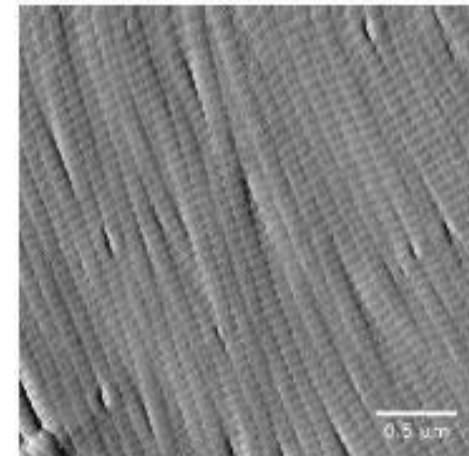
**Young skin (30yo)
Untreated – D8**



**Mature skin (65yo)
Untreated – D8**



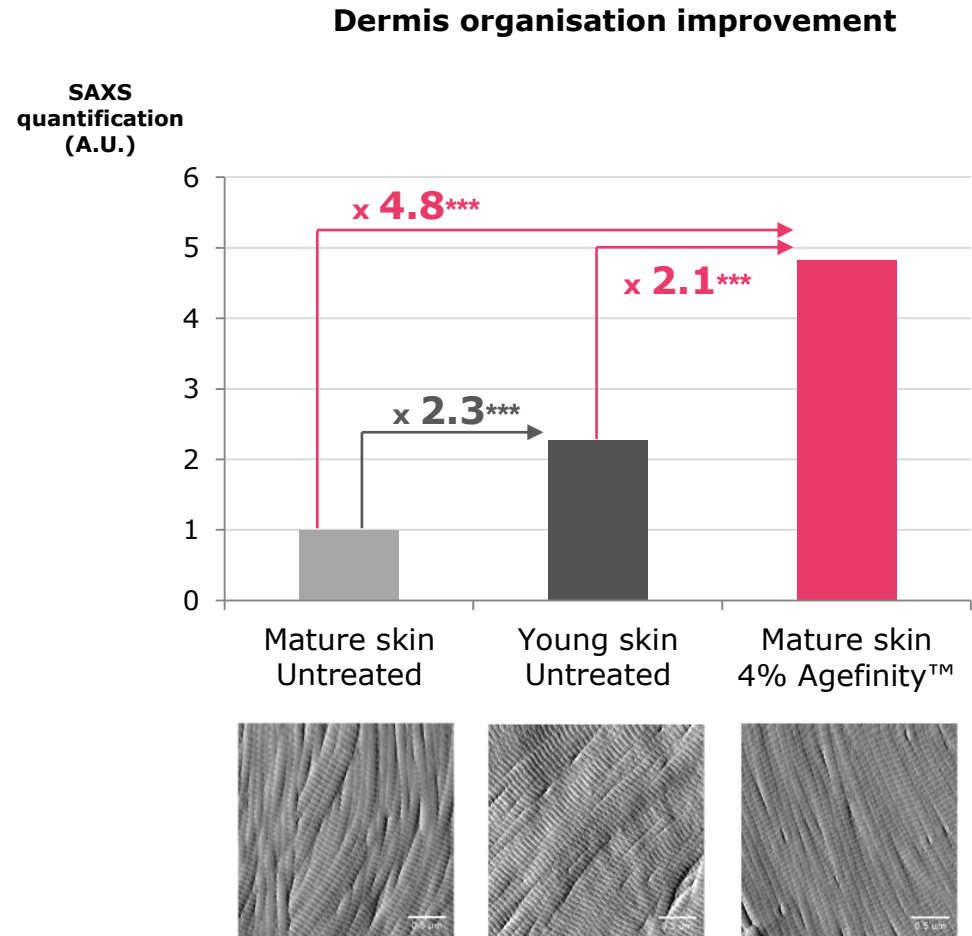
**Mature skin (65yo)
4% Agefinity™ – D8**



Collagen fibres organisation (*ex vivo*)

Results:

Agefinity™ at 4% **improves collagen fibres cohesion and organises the dermis** up to **4.8 times** better than untreated mature skin, and up to **2.1 times** better than untreated young skin.



Stimulation of skin detox.

Increase of proteasome activity (*ex vivo*)

Protocol:

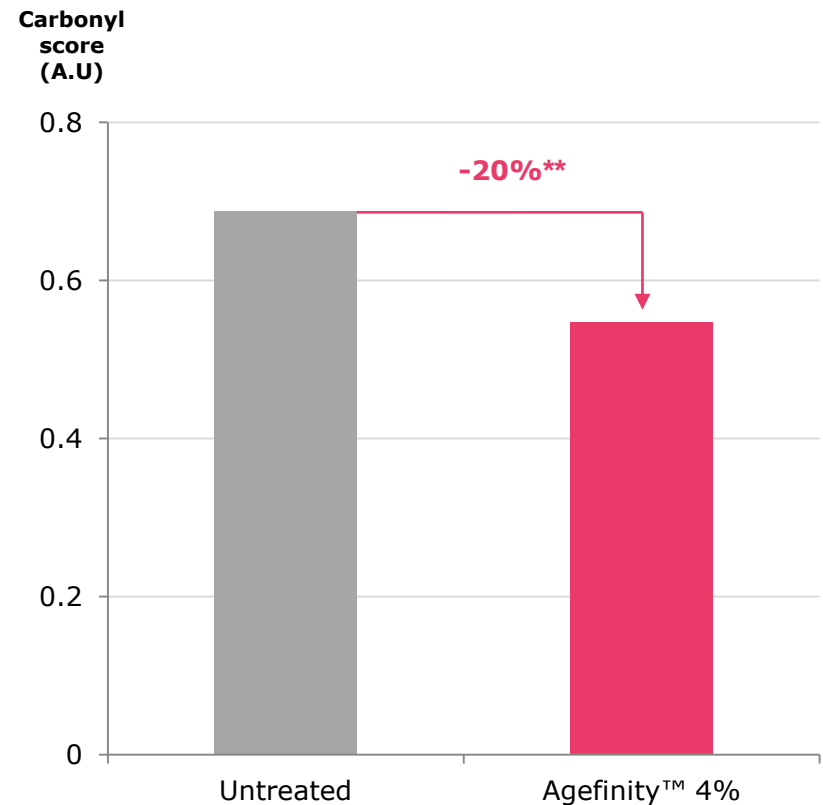
Skin explants from a 59 year old Caucasian woman were treated with 4% of Agefinity™ for 7 days or left untreated. The proteasome activity was evaluated by labelling carbonylated proteins (oxidised proteins) with a fluorescent probe.

Results:

Agefinity™ at 4% significantly reduces the carbonyl score by -20% compare to untreated reflecting a **significant reactivation of proteasome activity in a mature skin.**

****** $p < 0.01$ Student's *t*-test

Restoring proteasome activity (decrease of oxidised proteins content)



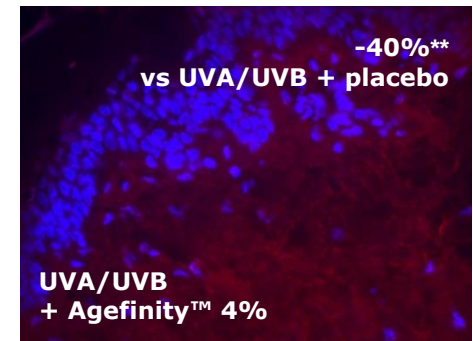
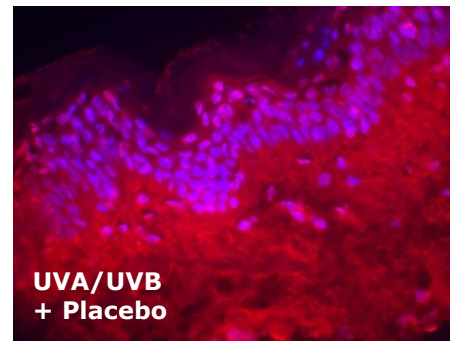
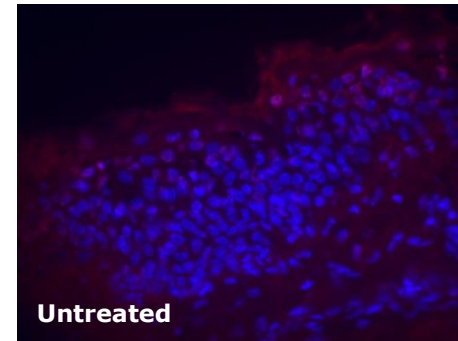
Reduction of oxidised proteins content (*ex vivo*)

Protocol:

Skin explants from a 45 year old Caucasian woman were pre treated with 4% of Agefinity™ or Placebo or left untreated for 24 hours. The skin explants were then exposed to UVA radiation (8J/cm²) and UVB radiations (200mJ/cm²) and treated again with the products for 24 hours. The protein oxidation level was then measured using immunostaining.

Results:

Agefinity™ at 4% **induces a significant reduction of oxidised proteins production under UV radiations**, down to -40% compared to the placebo.



** p<0.01 Student's t-test

Clinical efficacy

Anti-ageing efficacy (clinical)

Protocol n°1:

- 22 women (average age 59 years old)
- Crows feet wrinkles + Nasolabial wrinkles
- Twice a day
- 56 days
- Hemi-face: one side with a formula containing 4% Agefinity™ and the other side the Placebo.

For study (#1, #2 and #3)

INCI: AQUA/WATER, CETYL ALCOHOL, GLYCERYL STEARATE, PEG-75 STEARATE, CETETH-20, STEARETH-20, ISODECYL NEOPENTANOATE, ± **SODIUM MANNOS-6-PHOSPHATE**, PHENOXYETHANOL, METHYL PARABEN, PROPYL PARABEN, ETHYL PARABEN, DIMETHICONE, FRAGRANCE, BENZYL SALICYLATE, LINALOOL, D-LIMONENE.



1. Reduction of visible age spots

Protocol:

The number of visible spots was evaluated on using **VISIA CR 2.3®** to obtain **digital photography of the face at D0, D28 and D56.**

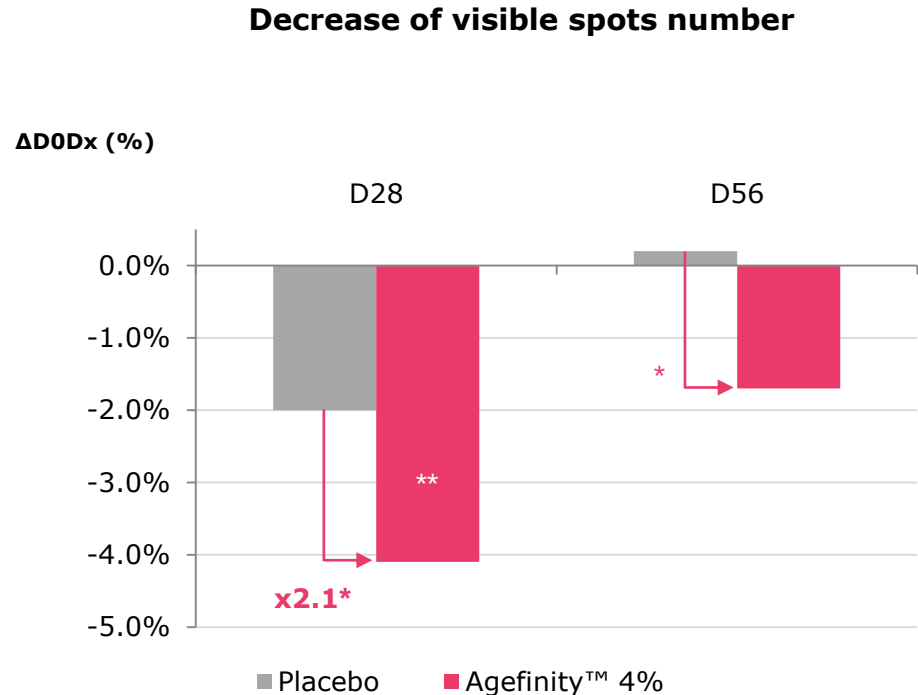


Visible reduction of age spots

Results:

After 28 days, Agefinity™ **significantly reduces the number of visible ageing spots** compared to placebo, with a reduction down to **2.1 times more efficient**.

The effect is still visible and significant versus placebo after 56 days.

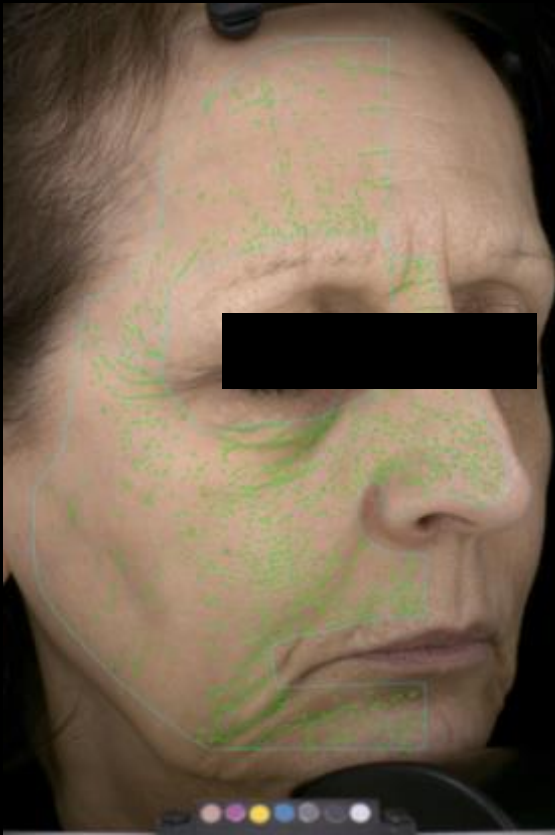


* $p < 0.05$ Student's t-test vs placebo

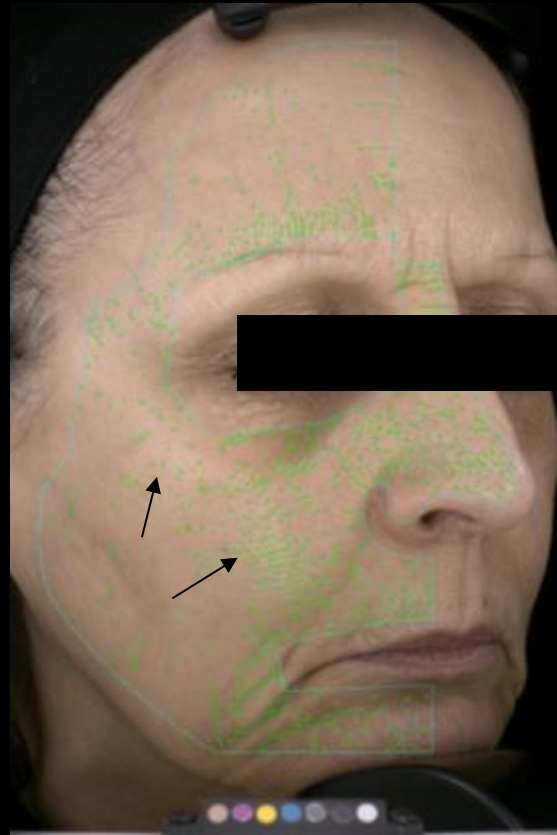
** $p < 0.05$ Student's t-test vs D0

Reduction of visible age spots

Visible spot with Agefinity™ at 4% (Vol GP180, 66 yo)



D0



D28



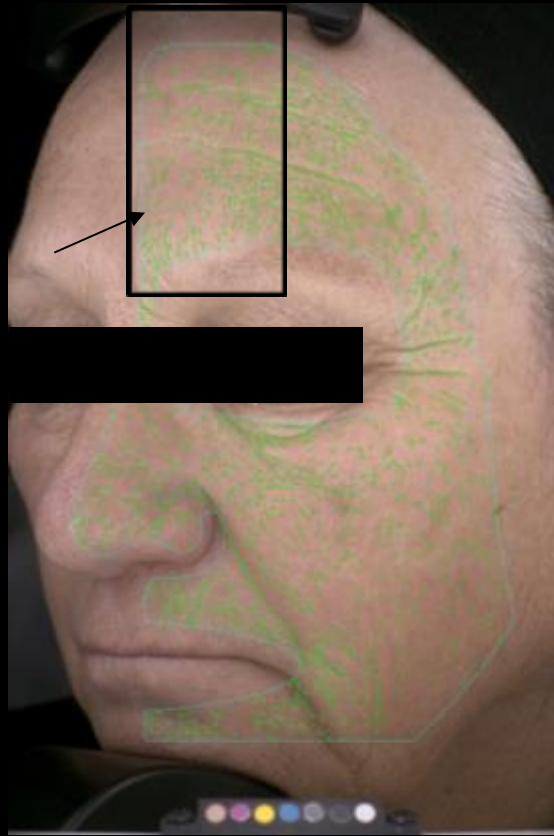
D56

Reduction of visible age spots

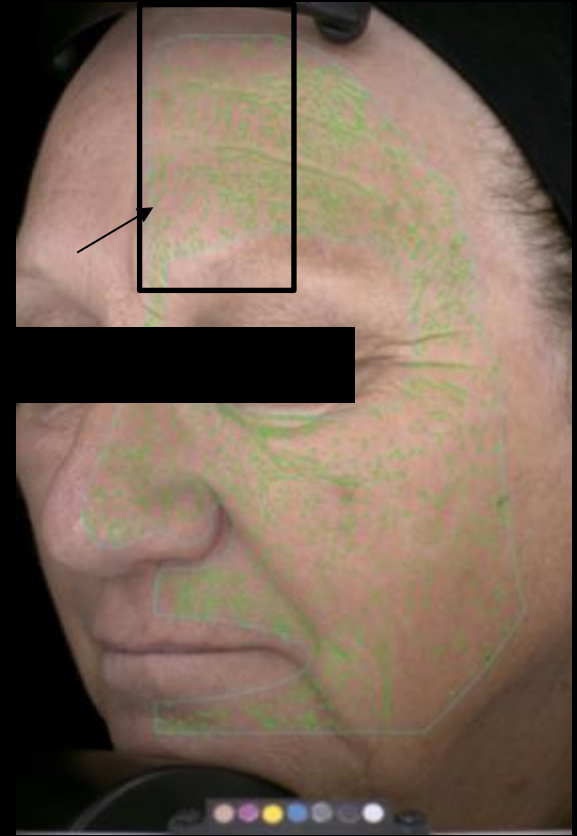
Visible spot with Agefinity™ at 4% (Vol GP178, 59 yo)



D0



D28



D56

2. Improvement of skin collagen organisation

Protocol:

The density and distribution of collagen were analysed at the commissure level using **SIAscope®** device and **SIAMetrics®** analysing system after 28 and 56 days.

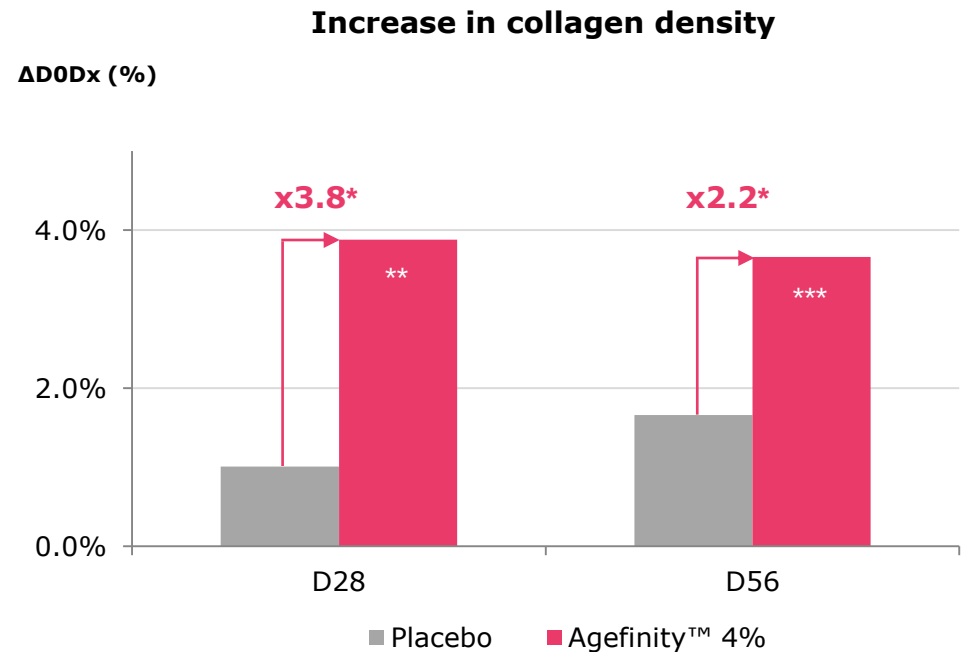
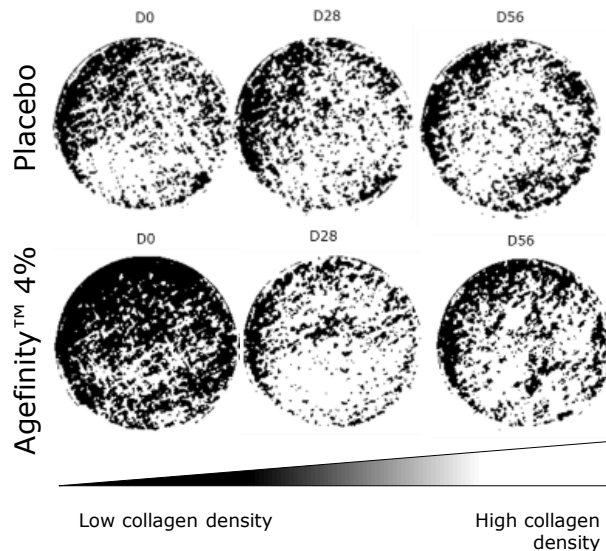


Improvement of collagen organisation

Results:

Agefinity™ **significantly stimulates relative collagen density** compared to placebo with an increase up to 3.8 folds.

The **collagen is better organised** after only 1 month.



* $p < 0.05$ Student's t-test vs Placebo

** $p < 0.01$ Student's t-test vs D0

*** $p < 0.001$ Student's t-test vs D0

3. Anti-wrinkle efficacy

Protocol:

The number of wrinkles was evaluated on the crow's feet area using **VISIA CR 2.3®** to obtain digital photography of the face at **D0, D28 and D56**.

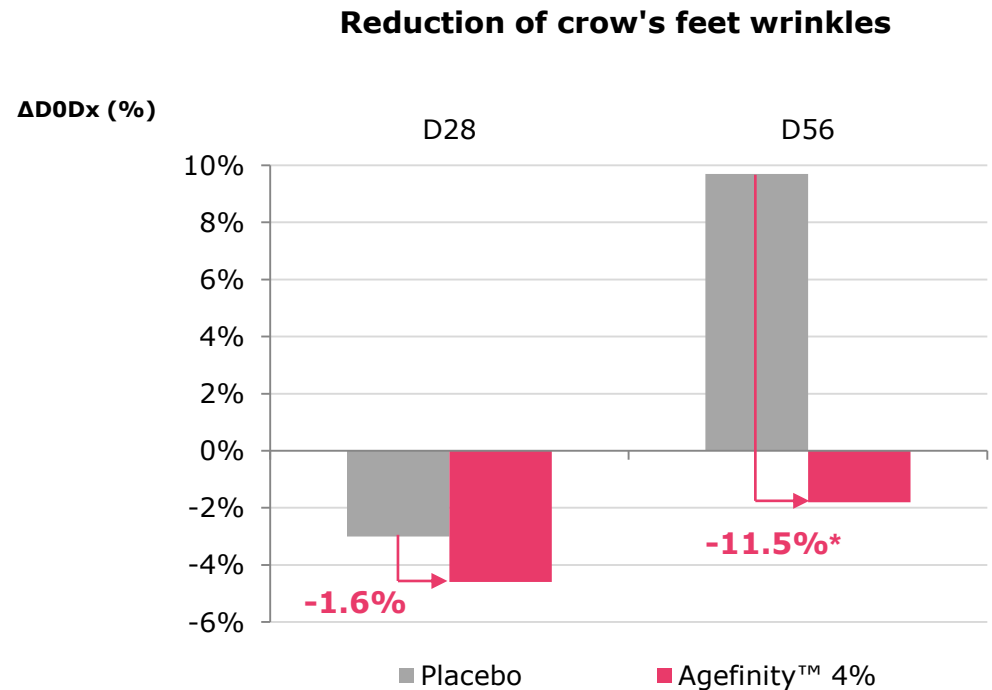


Significant reduction of crow's feet wrinkle number

Results:

Agefinity™ at 4% **induces a significant reduction of crow's feet wrinkle number**, down to -11.5%.

A visual improvement of the bottom part of the face (jowls) can also be noticed on the captured pictures.



* p<0.05 Wilcoxon test vs placebo

Anti-wrinkle efficacy

Crows' feet evaluation with Agefinity™ at 4% (Vol GPO38, 56 yo)



D0



D56

Anti-wrinkle efficacy

Crows' feet evaluation with Agefinity™ at 4% (Vol GP180, 66 yo)



D0



D56

Y-zone reshaper (clinical)

Protocol n°2:

- 20 women (average age 56 years old)
- Neck wrinkles
- Twice a day
- 56 days
- Full face: with a formula containing 4% Agefinity™ or a Placebo.

Neck wrinkles properties (volume and depth) were evaluated on the neck area using **AEVA HE®** at **D0, D28 and D56**.

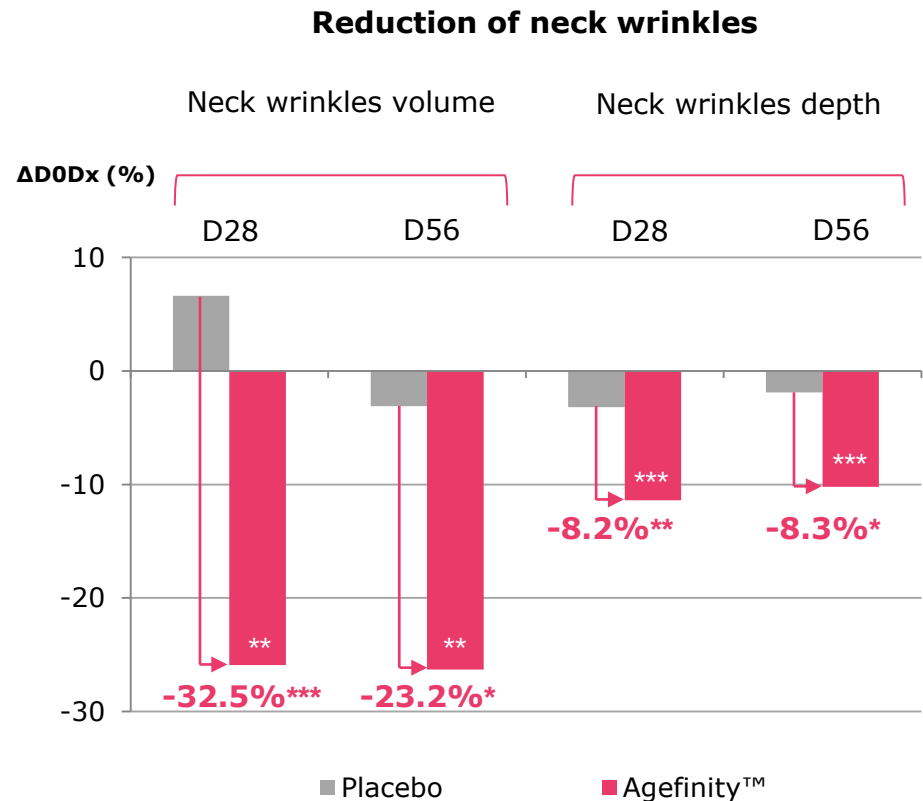
Panelists were also asked to assess both products efficacy on their neck after 56 days of use.



Reduction of the neck wrinkles volume and depth

Results:

Agefinity™ induces a significant **reduction of the neck wrinkles volume and depth**, respectively by up to **-32.5%** and **-8.3%** *versus* placebo.

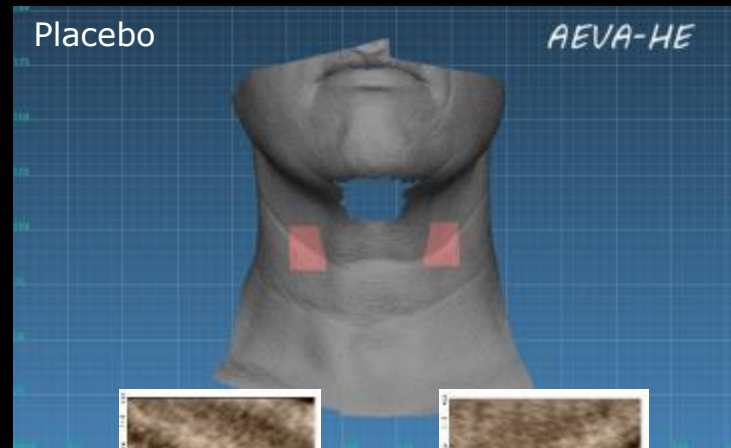


Wilcoxon test vs D0, Mann Whitney test vs placebo:
* p<0.05 ** p<0.01 *** p<0.001

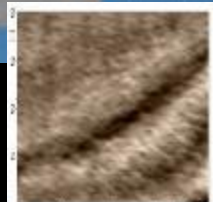
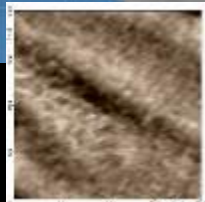
3D Y reshaper

Reduction of neck wrinkles with Agefinity™ 4% vs placebo

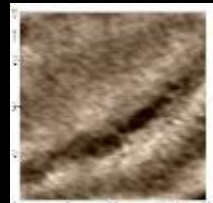
(Placebo: Vol 008, 46 yo, Agefinity™: Vol 273, 55 yo)



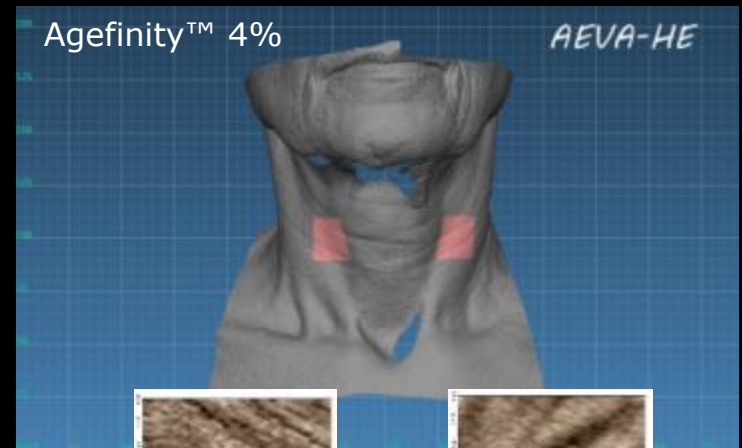
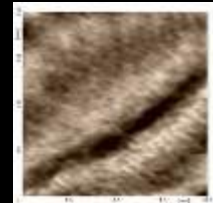
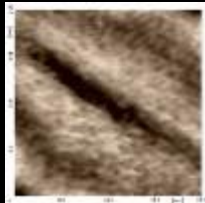
D0



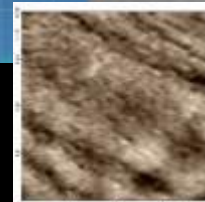
D28



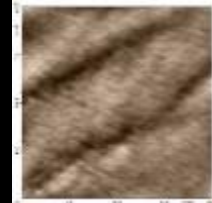
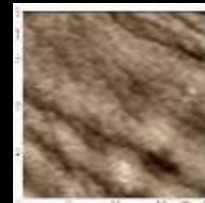
D56



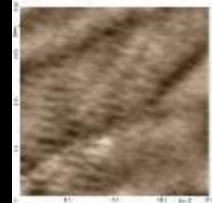
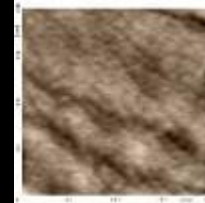
D0



D28



D56



Self assessment after 56 days of use

The skin of my neck
is firmer.

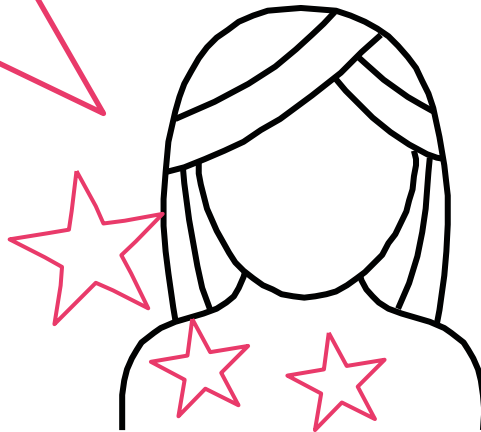
74%

(vs 45% for placebo)

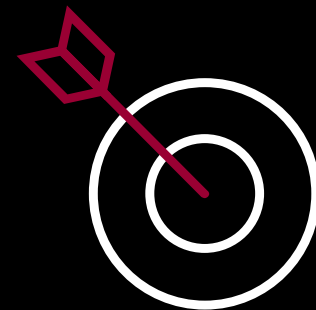
I'm convinced by the
effect of the product at
the end of this test.

84%

(vs 60% for placebo)

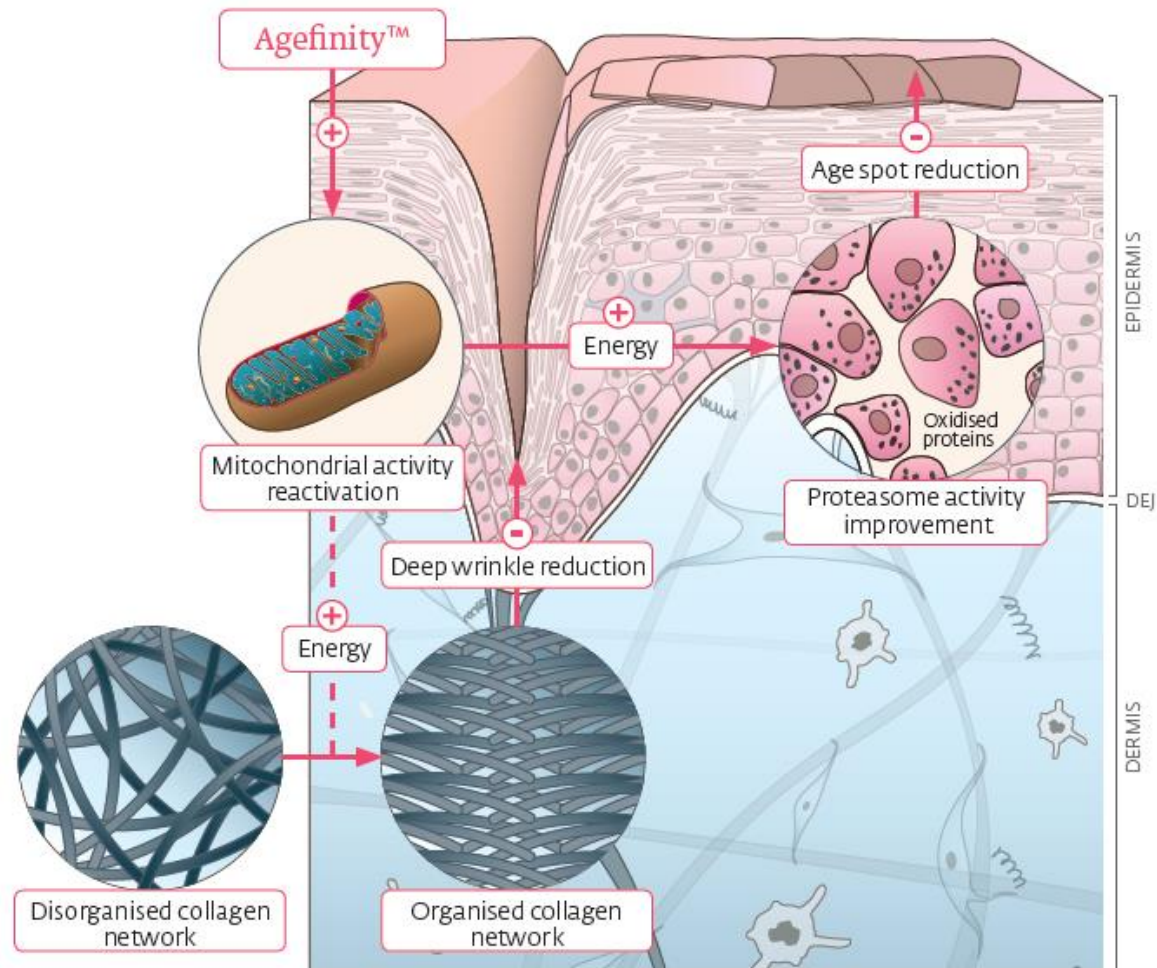


Summary



Agefinity™ mode of action

Recharge your skin!



Take home messages

Agefinity™ uniqueness

- Pure and clean source of **biomimetic energy for your skin**
- Relaunch mitochondrial activity, even in older cells
→ **Recharge your skin batteries!**
- Restructure reticular dermis
→ **“younger” collagen network**
(~35 years gain in 8 days observed under AFM)
- Reactivate skin detoxification
→ **Disappearance of age spots**
- Works on both face and neck
→ **visible reduction of wrinkles, including “tech” wrinkles**
- Patent-pending



Main claims

1. Anti-wrinkle
2. Collagen organiser
3. Skin lifting
4. Anti age-spots
5. Anti tech-neck
6. Remodeling the V-shape
7. Improvement of the Y-zone



Main applications

1. Anti-ageing serums or creams
2. Anti-ageing products for the reduction of deep wrinkles
3. Intensive treatments for facial rejuvenation
4. Nasolabial anti-wrinkles products
5. Anti tech-neck products
6. Anti-crow's feet products
7. Anti-age spots serums
8. Skincare for mature skin
9. Dermocosmetics



S3D® Mystic

Your eternal spark of youth

S3D® Mystic is a precious serum to use after your daily cleansing and before your day cream, or after your make-up removal and before your night cream. It features:

- **Agefinity™ (4%)**
A biomimetic source of energy which acts on the lower dermal layer reorganizing the skin matrix, visibly reducing age spots, crow's feet and neck wrinkles, to remodel the Y-shape of the face in 1 month.
- **Rose of Jericho Eutectys™ (2%)**
Coming from the desert, this "resurrection plant" is rich in phenolic acids and acts as skin anti-ageing.
- **Diamond powder (0.01%)**
It brings healing properties and stimulates immune defense system.



Agefinity™ - Rose of Jericho Eutectys™ BG - Diamond powder

Ref.: GSI 18-2

Phase	INCI name	Commercial name	Supplier	Dosage
A	Dimethicone, Dimethicone/PEG-10/15 copolymer	KSG-210	ShinEtsu	3%
	Dimethicone, Dimethicone/vinyl dimethicone crosspolymer	KSG-16	ShinEtsu	20%
	PEG-9 Polydimethylsiloxethyl dimethicone	KF-6028	ShinEtsu	1.5%
	Dimethicone	Xiameter PMX-200 Silicone Fluid 20 cSt	Dow corning	11.5%
	Fragrance, D-Limonene, Linalool	Diamond noir EAH64446/00	Givaudan	0.4%
B	Aqua/Water			22.09%
	Sodium Citrate	Sodium Citrate	Sigma	0.2%
	Sodium Chloride	Sodium Chloride	Brenntag	0.5%
C	Butylene glycol	Butylene glycol	Brenntag	24.5%
	Glycerin	Vegetal glycerin rapeseed	Oleon	10%
	Ethylhexylglycerin	Sensiva SC 50	Schülke	0.3%
D	Glycerin, Betaine, Aqua, Anastatica, Hierochuntica extract	Rose of Jericho Eutectys™ BG	Naturex	2%
	Glycerin, Water, Sodium mannose phosphate, Mannose	Agefinity™	Givaudan AB	4%
E	Diamond powder	Diamond powder	Givaudan AB	0.01%

Process

1. Prepare A under tooth propeller at room temperature until total homogenization
2. Prepare separately B, C and D under gentle stirring at room temperature
3. Add successively C and D in B under gentle stirring to total homogenization
4. Add very slowly BCD into A under stirring at 1500 rpm and increase the speed at 3500rpm until the gel is more viscous
5. Then add E.

Aspect: Opalescent gel • Viscosity (Brookfield Cylinder 4, Speed 12 at 20°C): 29000 mPa.s • pH : /

Technical information

Agefinity™

INCI

Glycerin (and) Water (and) Sodium Mannose Phosphate (and) Mannose

Origin

Plant & biotechnology

Preservation

Preservative free

Appearance

Clear liquid

Solubility

Hydrosoluble

Dosage

0.1% to 4%

Processing

Can be added to the water phase at the end of the formulation process, at a pH between 2 and 8.

Compliance



Thank you

Givaudan Active Beauty
global.cosmetic@givaudan.com